Università di Pisa
Department of Physics
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Anomalous Transport in Soaring Flights

Collective Strategies and Intermittent Search Models

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Chapter 1

Introduction

Chapter 2

The dataset

2.1 Source

The flights come from the *Coupe Fédérale de Distance* (CFD), the cross-country competition of the French free-flight federation (FFVL). Pilots declare their flights online, and each entry carries the flight's metadata and, in most cases, its GPS track. Paragliders and hang gliders have separate archives; a larger share of hang-glider entries has no track.

A season runs from 1 September to 31 August. We use the paraglider seasons from 1999–2000 and the hang-glider seasons from 2001–2002, up to 2025–2026. For each season we download the list of declared flights, kept verbatim as provenance, and the tracks it links. A saved list is never replaced, so each one is a snapshot of the day it was downloaded: the 2025–2026 lists were saved around the end of June 2026 and miss the rest of that season.

2.2 Tracks and metadata

Tracks are IGC files [1], plain-text records of the flight. Each `B` record is one *fix*: UTC time, latitude and longitude, a validity flag, and two altitudes, the pressure altitude from the barometer and the GNSS altitude from the satellite fix. The sampling interval depends on the recorder and need not be constant within a flight. Some recorders log no pressure altitude.

The metadata give the date, the pilot, the take-off and landing sites, the wing model and class, the scored distance and the scoring category of the declared route (free distance, out-and-return, flat or FAI triangle and a few others). The category describes the declared route only. A few paraglider entries have an incomplete date, such as `0000-00-00`, which excludes them from any comparison by date.

2.3 Glider categories

Wing design changes speed, glide performance and turning geometry, so paragliders and hang gliders are analysed separately [2]. A paraglider is a flexible canopy; a hang glider has a framed wing, flexible or rigid.

The wing class follows a different system in each discipline. Paraglider classes are those of the EN 926-2 standard, A to D, written with their equivalent in the older LTF scale ("C ou 2" reads "C or 2"), plus the CIVL Competition Class and a few other labels, such as tandem or uncertified wings. EN classes rank passive safety and the skill needed to recover from a collapse, and say nothing direct about speed or glide ratio [3]. Hang-glider classes are those of

Table 2.1. Paraglider flights per season.

Season	Flights	With track	Downloaded	Downloaded (% of tracks)
1999–2000	10	10	10	100.0
2000–2001	832	1	1	100.0
2001–2002	1,150	3	3	100.0
2002–2003	1,996	123	123	100.0
2003–2004	1,470	337	337	100.0
2004–2005	1,765	574	574	100.0
2005–2006	2,162	894	894	100.0
2006–2007	2,386	1,016	1,016	100.0
2007–2008	3,081	1,667	1,667	100.0
2008–2009	3,974	2,614	2,613	100.0
2009–2010	4,171	3,059	3,059	100.0
2010–2011	4,872	4,185	4,169	99.6
2011–2012	6,744	6,075	6,065	99.8
2012–2013	7,873	7,106	7,075	99.6
2013–2014	9,804	9,157	9,127	99.7
2014–2015	13,793	13,293	13,254	99.7
2015–2016	13,982	13,703	13,696	99.9
2016–2017	15,760	15,534	15,528	100.0
2017–2018	14,917	14,682	14,676	100.0
2018–2019	16,702	16,517	16,508	99.9
2019–2020	10,966	10,868	10,863	100.0
2020–2021	9,500	9,483	9,480	100.0
2021–2022	14,552	14,507	14,502	100.0
2022–2023	11,590	11,561	11,560	100.0
2023–2024	10,372	10,362	10,360	100.0
2024–2025	10,906	10,885	10,884	100.0
2025–2026	8,013	8,009	8,008	100.0
Total	203,343	186,225	186,052	99.9

Table 2.2. Hang-glider flights per season.

Season	Flights	With track	Downloaded	Downloaded (% of tracks)
2001–2002	248	0	0	–
2002–2003	281	6	6	100.0
2003–2004	256	28	25	89.3
2004–2005	244	35	25	71.4
2005–2006	365	80	76	95.0
2006–2007	337	100	96	96.0
2007–2008	388	136	132	97.1
2008–2009	344	155	155	100.0
2009–2010	324	136	136	100.0
2010–2011	425	147	147	100.0
2011–2012	117	103	103	100.0
2012–2013	428	406	406	100.0
2013–2014	423	409	409	100.0
2014–2015	628	595	591	99.3
2015–2016	417	413	413	100.0
2016–2017	468	462	462	100.0
2017–2018	342	331	331	100.0
2018–2019	604	594	594	100.0
2019–2020	378	373	373	100.0
2020–2021	328	328	328	100.0
2021–2022	570	568	568	100.0
2022–2023	376	374	374	100.0
2023–2024	437	437	436	99.8
2024–2025	353	352	352	100.0
2025–2026	178	178	178	100.0
Total	9,259	6,746	6,716	99.6

the FAI [4]: Class 1, with its Sport sub-class, for wings controlled by weight shift alone, and Classes 2 and 5 for rigid wings with movable control surfaces.

Chapter 3

Trajectory pre-processing

The pre-processing turns each IGC file into a trajectory fit for analysis. This chapter describes its steps in the order they run, one section per step. Some steps delete single fixes; others judge the whole flight and keep or drop it.

3.1 Reading the fixes

Each `B` record gives a fix $(t_i, \varphi_i, \lambda_i)$, with latitude and longitude in degrees, together with the validity flag and the two altitudes h_i^{baro} and h_i^{GNSS}. A record that does not follow the fixed layout of the format, or whose time of day or position is impossible, is discarded. A missing or unreadable altitude, including the zero the format uses for a missing reading [1], is marked as missing, and the fix keeps its time and position.

Time is measured in seconds from the first fix. Records carry only the time of day, so a flight crossing UTC midnight would jump back by a day; a day is added when the clock passes from the last hour of one day to the first hour of the next. Other backward steps and repeated times are left to the clock rule of Section 3.3. Well-formed but implausible fixes, such as position spikes, are kept as well: the parser discards only what cannot be read. The validity flag, which marks fixes without a three-dimensional GNSS solution, is kept but not used at this stage.

3.2 Choice of altitude channel

Some recorders log no pressure altitude, so we use GNSS altitude for every flight, $z = h^{\text{GNSS}}$. Its vertical datum is not uniform across recorders: the IGC specification prescribes height above the WGS84 ellipsoid [1], the CIVL specification for hang gliders and paragliders height above the geoid [5]. A constant offset cancels in altitude increments, but comparing absolute altitudes across flights would need a datum audit.

Let N be the number of fixes read from the file, before any is deleted, and P the set of those with a GNSS altitude. The flight is kept only if

$$\frac{|P|}{N} \geq 0.95 \qquad \text{and} \qquad \max_{i \in P} h_i^{\text{GNSS}} - \min_{i \in P} h_i^{\text{GNSS}} \geq 30\,\text{m}. \tag{3.1}$$

The first condition rejects incomplete channels; the second rejects channels stuck at one value, and flights with almost no vertical excursion. Fixes flagged as lacking a three-dimensional solution count as present if they carry an altitude. Pressure altitude is never substituted for a missing GNSS value.

The barometer is checked with the same two conditions on h^{baro}, with thresholds 0.95 and 30 m. A flight whose barometer passes is marked, so that later cleaning steps can use pressure altitude as independent evidence; the mark does not decide whether the flight is kept.

3.3 Clock

A recorder can write a time equal to or earlier than the one before it. On the fixes as read, the clock rule keeps the first fix, and fix i only if its time is later than that of every fix before it,

$$t_i > \max_{j<i} t_j. \tag{3.2}$$

Of several fixes with the same second, the first is kept. A backward step is deleted, and so is every fix after it until the clock passes the latest time already kept. The times left are strictly increasing, so every step between consecutive fixes has a positive duration (Section 3.5).

3.4 Repeated positions

Some recorders repeat their last position while the clock runs, so a fix can carry the coordinates of the one before it without being a new measurement. On the fixes left by the clock rule, numbered again in order, fix i is deleted if

$$\varphi_i = \varphi_{i-1} \qquad \text{and} \qquad \lambda_i = \lambda_{i-1}. \tag{3.3}$$

The comparison is exact: the two records carry the same digits. Altitudes are not compared. In a run of fixes at the same position only the first is kept.

3.5 Fast tracks

A free flight spends part of its time climbing in thermals at a low ground speed, so the typical speed of its track stays well below the top speed of the wing. A track with a higher typical speed comes from something else, such as a powered aircraft or a file with compressed times, and the whole flight is dropped.

Let n be the number of fixes left by the two previous rules. Between fix i and fix $i+1$ the horizontal distance is taken on a sphere of radius $R = 6{,}371$ km, with the haversine formula

$$d_i = 2R \arcsin \sqrt{\sin^2 \frac{\varphi_{i+1} - \varphi_i}{2} + \cos\varphi_i \cos\varphi_{i+1} \sin^2 \frac{\lambda_{i+1} - \lambda_i}{2}}, \tag{3.4}$$

with the angles in radians, and the step speed is

$$v_i = \frac{d_i}{t_{i+1} - t_i}, \qquad i = 1, \ldots, n-1, \tag{3.5}$$

. The clock rule makes every denominator positive. The typical speed of the flight is the median of its step speeds,

$$\tilde{v} = \mathrm{median}\{v_1, \ldots, v_{n-1}\}, \tag{3.6}$$

and the flight is kept if $\tilde{v}$ does not exceed 70 km/h for paragliders and 110 km/h for hang gliders. A flight with fewer than two fixes has no step and is kept. The rule deletes no fix, and the median is not moved by a few wrong steps, such as a position spike: those are left to a later step.

The thresholds were read off the distribution of $\tilde{v}$ over all flights (Figure 3.1). In both disciplines the bulk of the distribution ends below the threshold, and beyond it only a sparse tail of single flights remains. Split by wing class (Figures 3.2 and 3.3), the bulk of every class stays below the threshold. The faster shoulder of the hang-glider distribution comes from the rigid Class 2 wings.

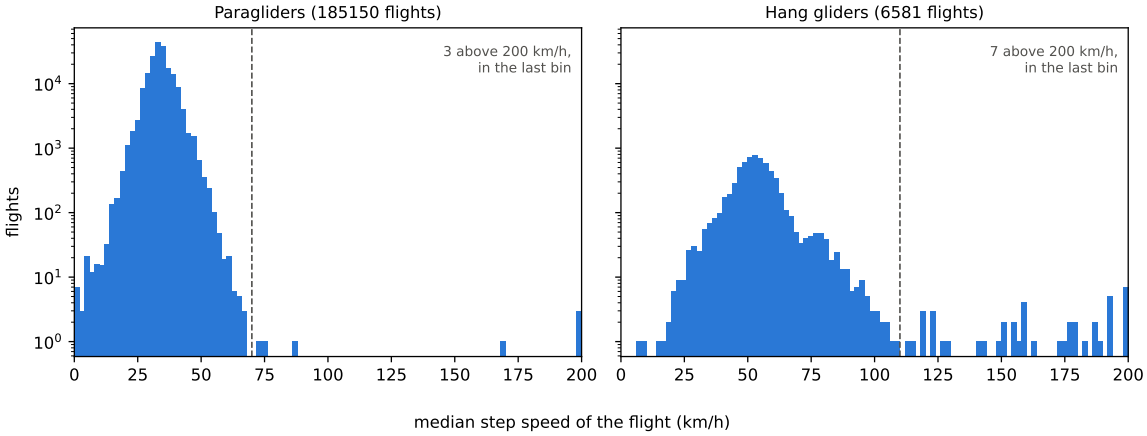


Figure 3.1. Median step speed $\tilde{v}$ (Equation (3.6)) of every flight past the altitude channel, computed after the clock and repeated-position rules. Bins of 2 km/h, logarithmic vertical axis; flights above 200 km/h fall in the last bin. Dashed line: threshold, 70 km/h for paragliders and 110 km/h for hang gliders.

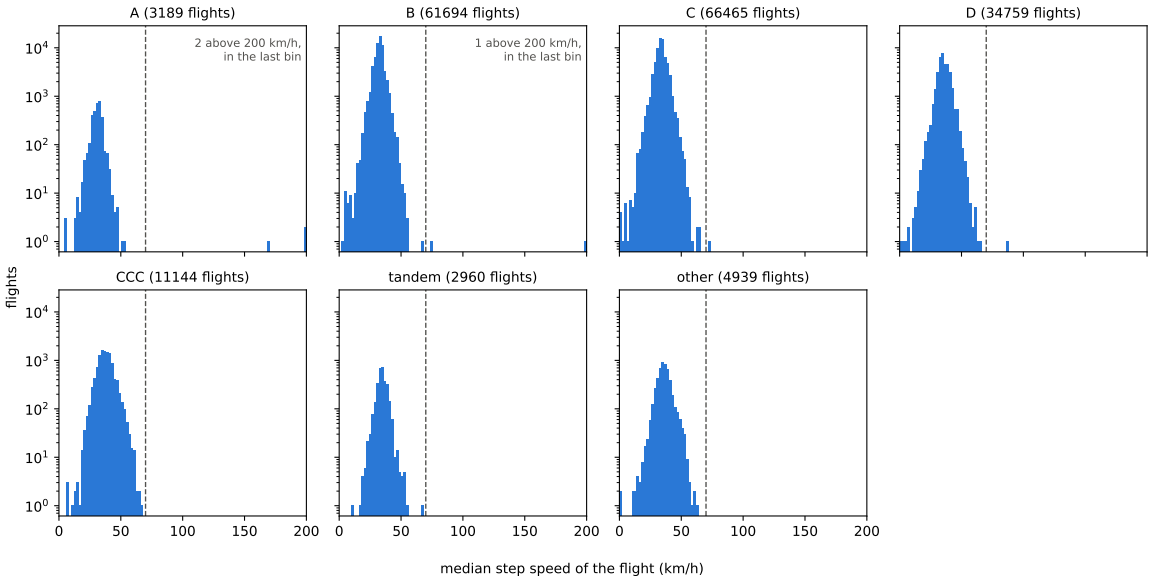


Figure 3.2. As Figure 3.1, for paragliders by wing class (Section 2.3): EN classes A to D, the CIVL Competition Class (CCC), tandem wings, certified or not, and every other label.

3.6 Horizontal excursions

A step faster than any glider of the discipline can fly is a recorder error. The excursion rule uses a top speed $v_{\max}$, read off the data at the end of this section, to delete the fixes such steps lead to, or to split the flight where the track does not come back.

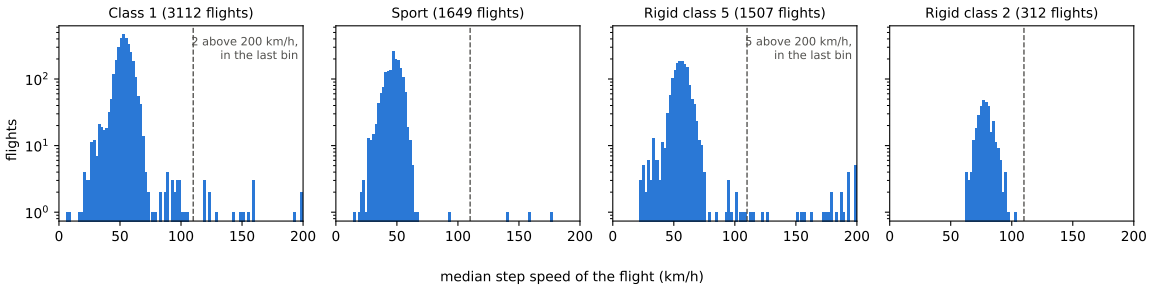


Figure 3.3. As Figure 3.1, for hang gliders by wing class (Section 2.3): Class 1, its Sport sub-class, and the rigid Classes 5 and 2. Flights of any other class are not shown.

The rule scans the fixes left by the previous steps in order, against an anchor a, the last fix accepted. Fix j is reachable from the anchor if

$$d(a, j) \leq v_{\max}\,(t_j - t_a), \tag{3.7}$$

with $d(a, j)$ the distance of Equation (3.4) between the two fixes. A reachable fix is accepted and becomes the anchor. An unreachable fix j opens a block, and the rule looks for a later fix k reachable from the anchor within $W = 20\,\text{s}$ of the opening, $t_k - t_j \leq W$. If there is one, the track went out and came back: fixes j to $k-1$ are deleted and k becomes the anchor. If there is none, the track does not come back: the flight is split between the anchor and fix j, and a new segment starts from j. A block still open at the end of the track is deleted.

The first anchor of each segment, at the start of the track or after a split, is the first fix whose step to the next fix satisfies Equation (3.7); the fixes before it are deleted. An anchor in the wrong place would make every good fix after it unreachable, while a wrong fix agrees with neither of its neighbours and so never becomes an anchor. The price is that a good fix followed by a spike is deleted with it. The rule leaves the fixes it accepted and the places where the flight is split, which a later step uses as the boundaries of its segments.

The step speed of Equation (3.5) mixes real flight with recorder errors, whose steps can be arbitrarily fast. Over a window of length W the speed from fix i is

$$u_i = \frac{d(i, j)}{t_j - t_i}, \qquad j = \min\{k :\ t_k - t_i \geq W\}, \tag{3.8}$$

with $d(i, j)$ as in Equation (3.7). Across a gap the window is longer than W. A single fix displaced by a distance D changes u_i by at most D/W, against $D/\Delta t$ for the two steps that touch it, so the window speed follows real flight much further than the step speed (Figure 3.4, lower row). Both distributions fall steeply; the step speed then turns into a long tail of errors, the window speed only later and much lower.

The window speed still carries errors that last longer than W, such as a jump that never comes back. Split by wing class (Figure 3.4, upper row), real flight depends on the wing: each class falls off in the order of its performance. The errors do not depend on the wing, and beyond the fall-off the curves of all classes merge into one common tail. Real flight therefore ends where the fastest class reaches that tail. We set $v_{\max}$ there for paragliders, 100 km/h, and with a margin above the end of the rigid Class 2 curve for hang gliders, 175 km/h.

The window lasts as long as the time the excursion rule waits for a track to come back. The rule splits a flight only where the mean speed from the last accepted fix stays above $v_{\max}$ for W, so a threshold above every real window speed never splits real flight.

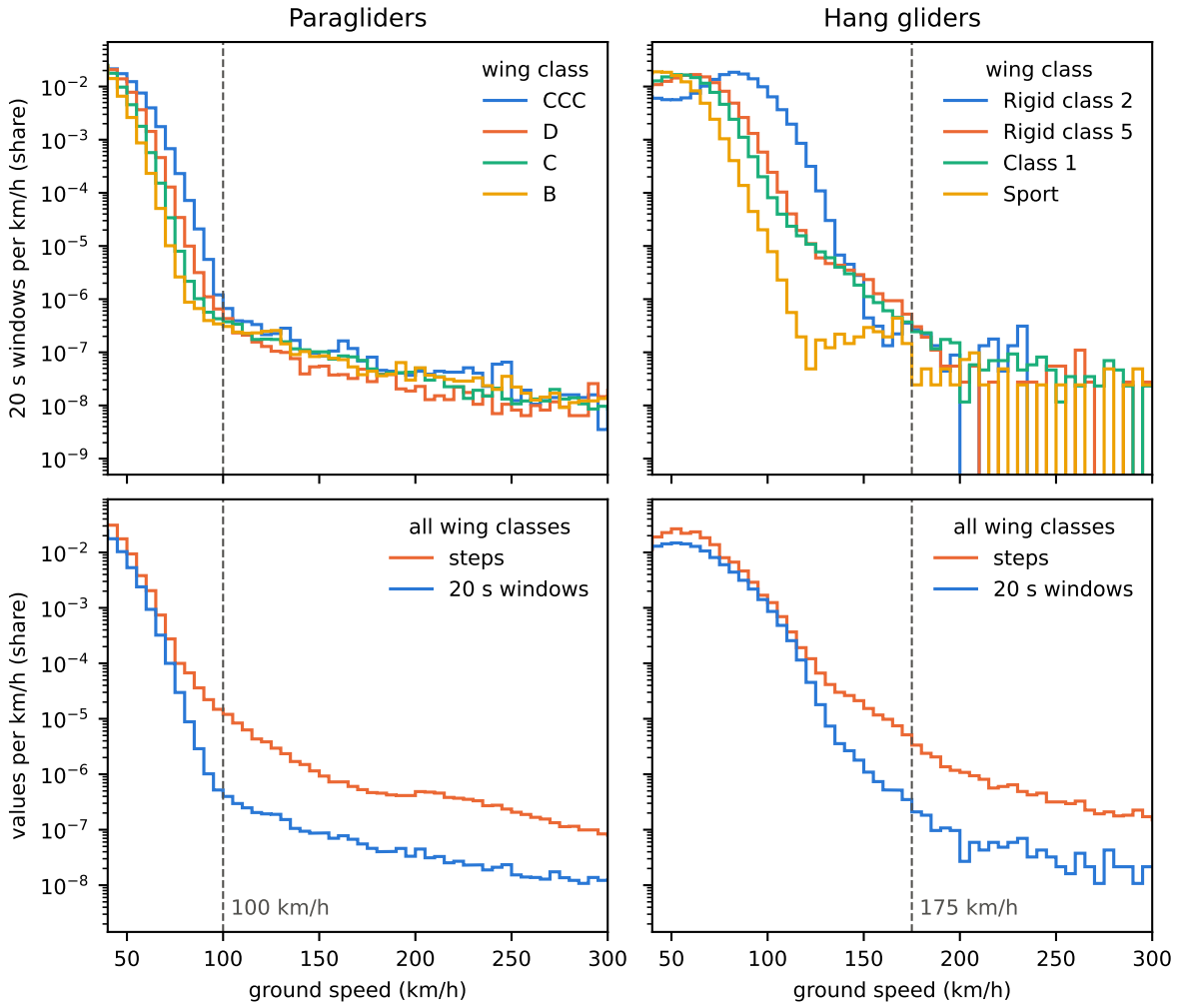


Figure 3.4. Ground speed in the flights kept by the previous steps. Upper row: speed over windows of 20 s, Equation (3.8), for four wing classes of each discipline (Section 2.3). Lower row, all classes together: speed of single steps, Equation (3.5), and over windows. Each curve is the share of its values per km/h, in bins of 5 km/h, on a logarithmic axis; values beyond the axis count in the share. Dashed line: the top speed $v_{\max}$, 100 km/h for paragliders and 175 km/h for hang gliders.

Bibliography

[1] Fédération Aéronautique Internationale, International Gliding Commission. Technical specification for IGC-approved GNSS flight recorders. Second Edition, with Amendment 8, 2023. Defines the IGC flight-data file format. https://www.fai.org/sites/default/files/igc_fr_specification_with_al8_2023-2-1_0.pdf.

[2] Jérémie Vilpellet, Alexandre Darmon, and Michael Benzaquen. From random walks to thermal rides: Universal anomalous transport in soaring flights. arXiv:2601.01293v1 [cond-mat.stat-mech], 2026. Version 1, 3 January 2026.

[3] CEN/TC 136/WG 6. Paragliding equipment — paragliders — part 2: Requirements and test methods for classifying flight safety characteristics. Technical Report Draft of FprEN 926-2, document N 0054, submitted to formal vote, European Committee for Standardization, 2012. Final draft of the standard published as EN 926-2:2013. http://www.xcmag.com/wp-content/uploads/2012/08/DraftEN926-2.pdf.

[4] Fédération Aéronautique Internationale. FAI sporting code, common section 7: Hang gliders and paragliders. 2024 edition, 2024. Defines the FAI hang-glider classes and their subclass ceilings (§1.4.1). https://www.fai.org/sites/default/files/civl/documents/sporting_code_s7_-_common_2024_v3.pdf.

[5] Fédération Aéronautique Internationale, CIVL. Sporting code section 7h: Civl flight recorder specification. 2024 edition, stated effective date 1 May 2024, 2024. Section 3.2.1 specifies GNSS altitude relative to the geoid. The available document retains draft annotations. https://fai.org/sites/default/files/civl/documents/sporting_code_s7_h_-_civl_flight_recorder_specification_2024.pdf.